

Aim

To create a Kafka Stream application that reads messages from an existing Kafka topic and displays them.

Prerequisites

Java JDK 8 or later

Apache Kafka installed

Existing Kafka topic (e.g., student)

Step 1: Start Kafka

Open separate Command Prompts.

Start ZooKeeper (if using older Kafka versions):

```
bin\windows\zookeeper-server-start.bat config\zookeeper.properties
```

Start Kafka Broker:

```
bin\windows\kafka-server-start.bat config\server.properties
```

Step 2: Create a Topic (if it doesn't already exist)

```
bin\windows\kafka-topics.bat --create --topic student --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
```

Step 3: Send Messages

Start the producer:

```
bin\windows\kafka-console-producer.bat --topic student --bootstrap-server localhost:9092
```

Enter messages:

hello

this is second Pract

Step 4: Create a Kafka Streams Application

Write Java Code

Step 5: Run the Application

Run the Java program.

```
import org.apache.kafka.common.serialization.Serdes;
import org.apache.kafka.streams.KafkaStreams;
import org.apache.kafka.streams.StreamsBuilder;
import org.apache.kafka.streams.StreamsConfig;
import org.apache.kafka.streams.kstream.KStream;
import java.util.Properties;

public class KafkaStreamExample {
```

```

public static void main(String[] args) {
    Properties props = new Properties();
    props.put(StreamsConfig.APPLICATION_ID_CONFIG, "student-stream-app");
    props.put(StreamsConfig.BOOTSTRAP_SERVERS_CONFIG, "localhost:9092");
    props.put(StreamsConfig.DEFAULT_KEY_SERDE_CLASS_CONFIG,
        Serdes.String().getClass());
    props.put(StreamsConfig.DEFAULT_VALUE_SERDE_CLASS_CONFIG,
        Serdes.String().getClass());
    StreamsBuilder builder = new StreamsBuilder();
    KStream<String, String> stream = builder.stream("student");
    stream.foreach((key, value) ->
        System.out.println("Received: " + value));
    KafkaStreams streams = new KafkaStreams(builder.build(), props);
    streams.start();
    Runtime.getRuntime().addShutdownHook(new Thread(streams::close));
}
}

```

```

PS C:\kafka_2.12-3.7.0> bin\windows\kafka-console-producer.bat --topic student --bootstrap-server localhost:9092
hello
this is second pract

streams.KafkaStreams - stream-client [student-stream-app-eeed3288-b742-4611-a66b-23]
ALANCING to RUNNING
2026-08-12 09:03:04,998 [student-stream-app-eeed3288-b742-4611-a66b-23]
clients.consumer.internals.LegacyKafkaConsumer - [Consumer clientId=student-stream-app-207-StreamThread-1-consumer, groupId=student-stream-app] Requesting the
lag
Received: hello
Received: this is second pract

PS C:\KafkaPrograms> javac -cp "C:\kafka_2.12-3.7.0\libs\*" KafkaStreamExample.java
PS C:\KafkaPrograms> java -cp ".;C:\kafka_2.12-3.7.0\libs\*" KafkaStreamExample
2026-08-12 09:03:04,439 [main] INFO org.apache.kafka.streams.StreamsConfig - StreamsConfig values:
    acceptable.recovery.lag = 10000
    application.id = student-stream-app
    application.server =
    auto.include.jmx.reporter = true
    bootstrap.servers = [localhost:9092]
    buffered.records.per.partition = 1000
    built.in.metrics.version = latest
    cache.max.bytes.buffering = 10485760
    client.id =
    commit.interval.ms = 30000
    connections.max.idle.ms = 540000

```